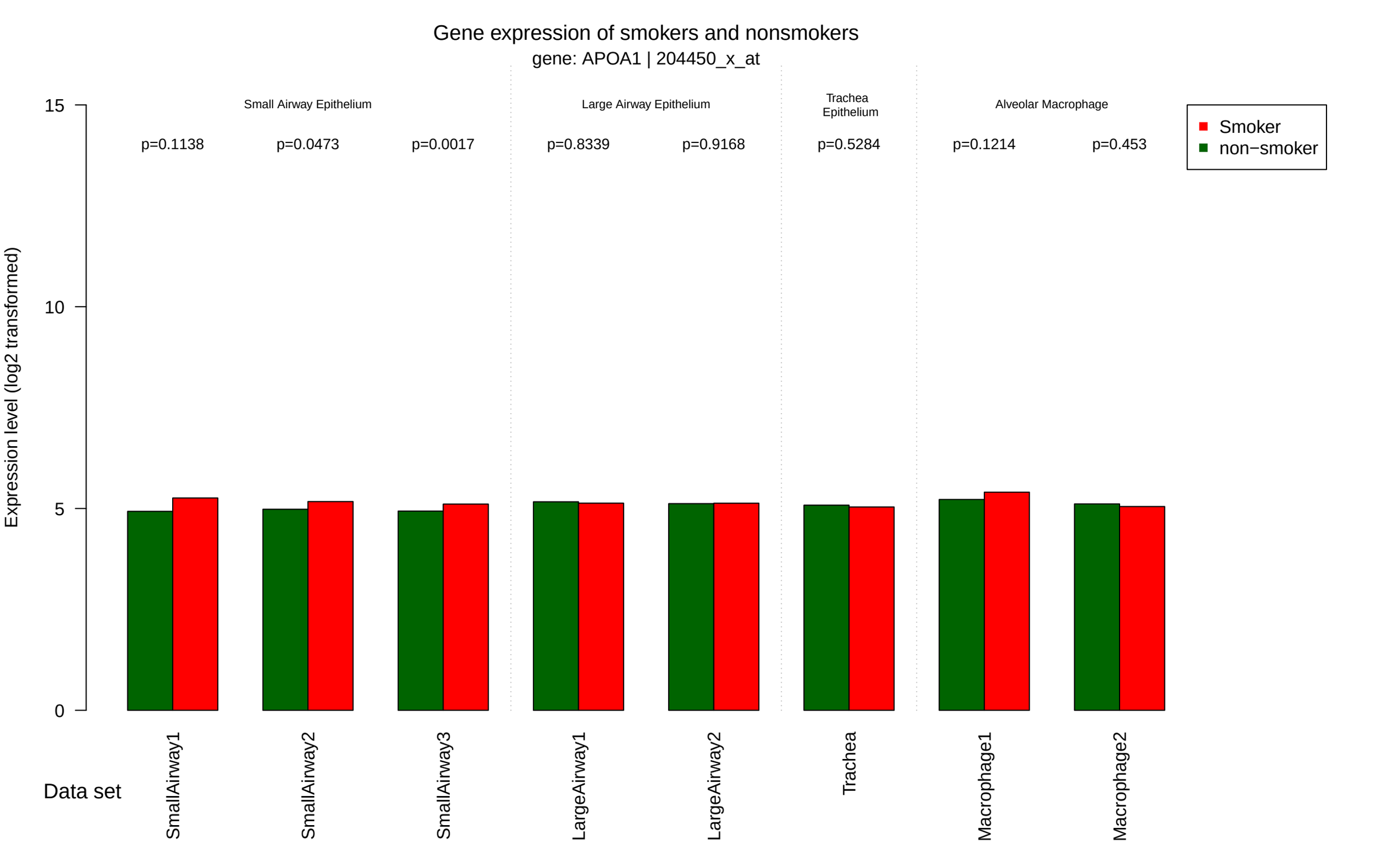
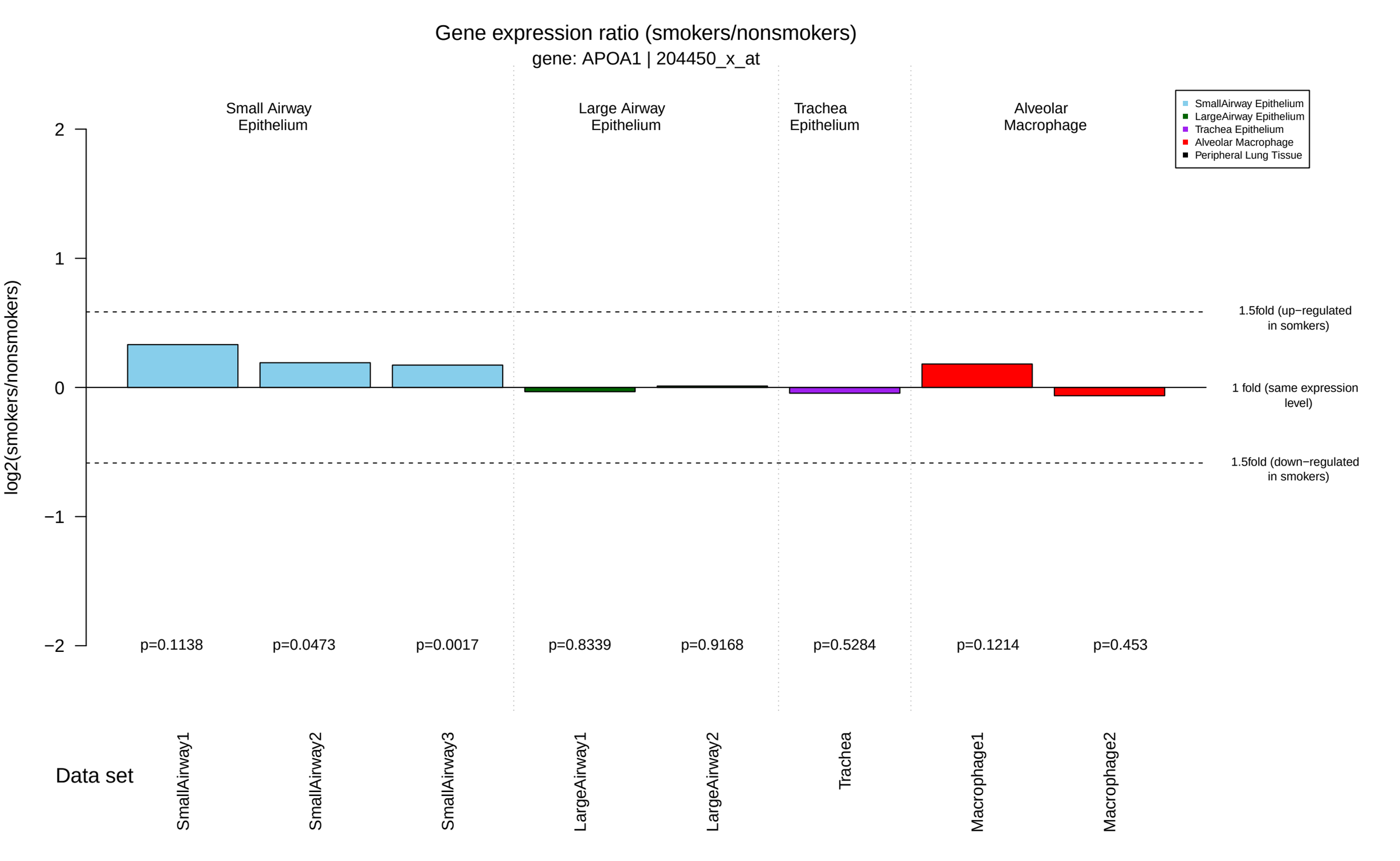
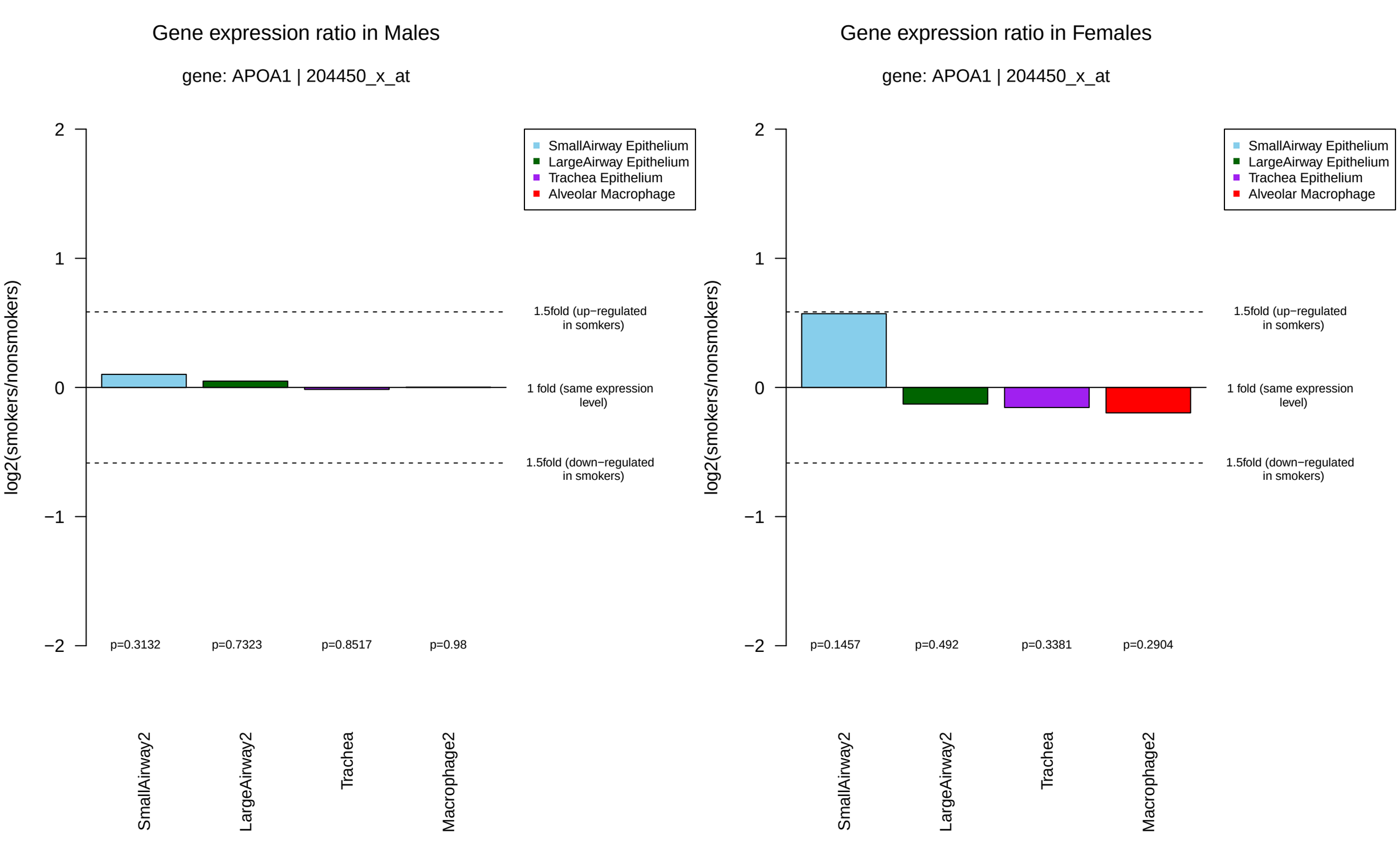
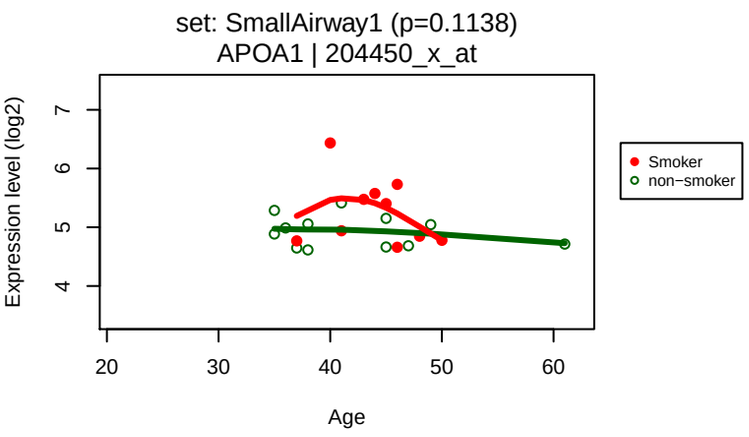
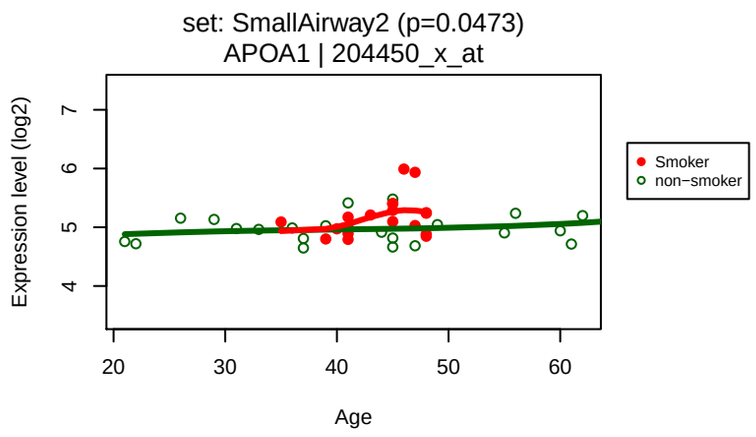
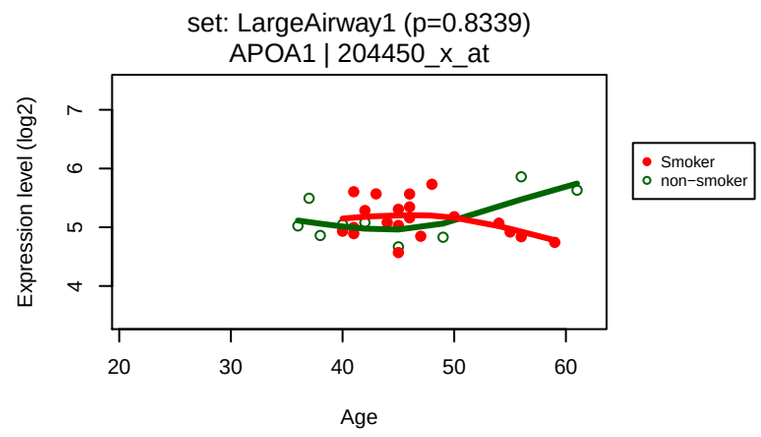

Gene expression of smokers and nonsmokers
gene: APOA1 | 204450_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1138
p=0.0473
p=0.0017
p=0.8339
p=0.9168
p=0.5284
p=0.1214
p=0.453
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

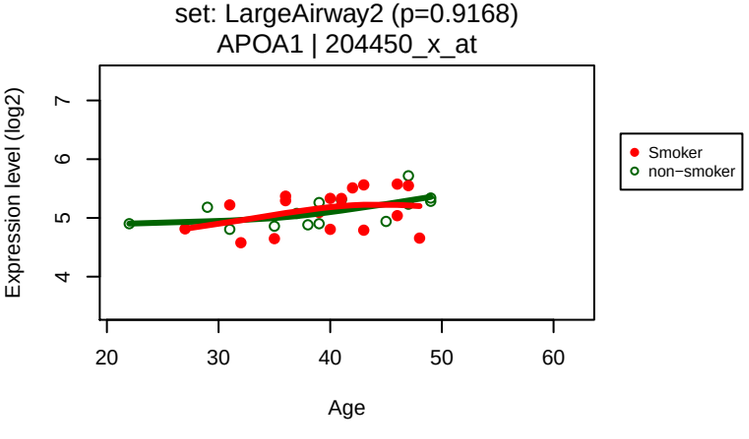

Gene expression ratio (smokers/nonsmokers)
gene: APOA1 | 204450_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1138
p=0.0473
p=0.0017
p=0.8339
p=0.9168
p=0.5284
p=0.1214
p=0.453
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

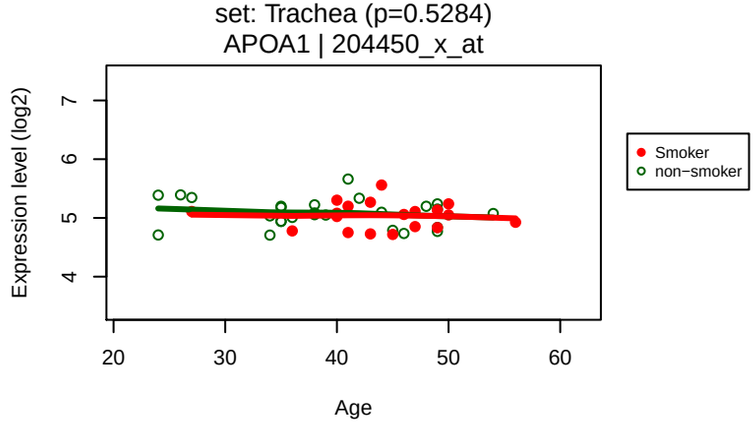

Gene expression ratio in Males
gene: APOA1 | 204450_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3132
p=0.7323
p=0.8517
p=0.98
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: APOA1 | 204450_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1457
p=0.492
p=0.3381
p=0.2904
SmallAirway2
LargeAirway2
Trachea
Macrophage2

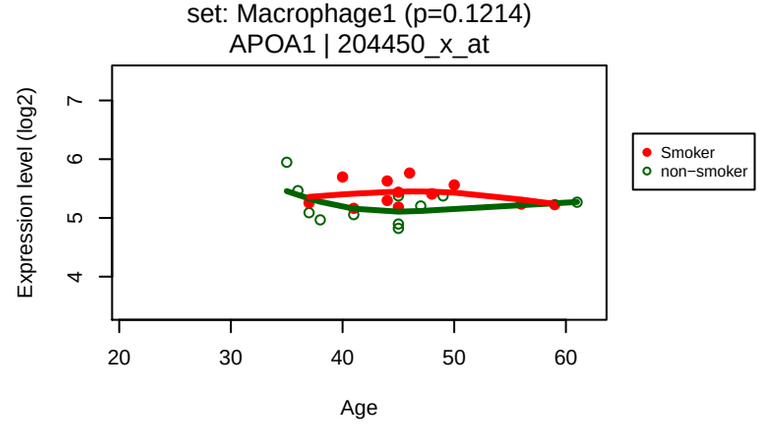

set: SmallAirway1 (p=0.1138)
APOA1 | 204450_x_at
Expression level (log2)
Age
Smoker
non−smoker

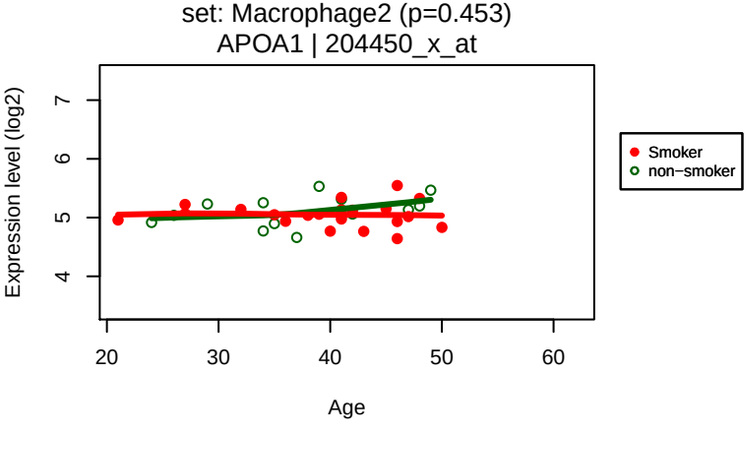

set: SmallAirway2 (p=0.0473)
APOA1 | 204450_x_at
Expression level (log2)
Age
Smoker
non−smoker

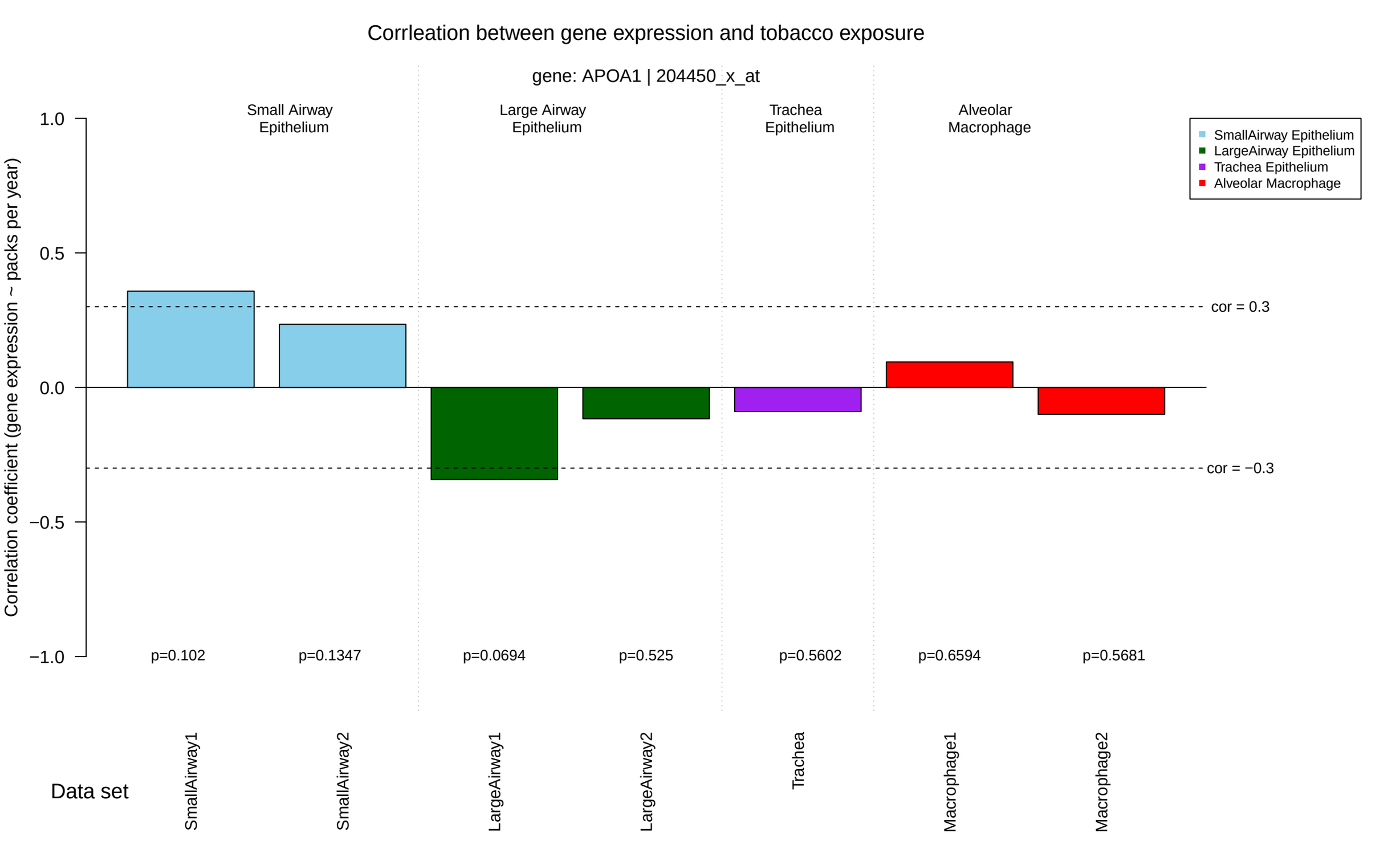

set: LargeAirway1 (p=0.8339)
APOA1 | 204450_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9168)
APOA1 | 204450_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5284)
APOA1 | 204450_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1214)
APOA1 | 204450_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.453)
APOA1 | 204450_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: APOA1 | 204450_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.102
p=0.1347
p=0.0694
p=0.525
p=0.5602
p=0.6594
p=0.5681
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2